Measure theory in a nutshell

- A measure μ assigns mass to subsets of a space: $\mu(A) \in \mathbb{R} \cup \{-\infty, \infty\}$
- Non-negative measure: $\forall A \subset X \quad \mu(A) \geq 0$
- Probability measure: $\mu(X) = 1$
- Like a "generalized histogram" — works for discrete or continuous data
- A measure lives on a *measurable space* $(\mathscr{X}, \Sigma)$, where Σ is a collection of subsets we're allowed to assign mass to (σ-algebra)

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